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ANALYSIS OF TRANSIENT STABILITY IN A PRACTICAL POWER SYSTEM UNDER DIFFERENT CONTINGENCIES

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ABSTRACT

An essential phase in power system analysis is the investigation of an electric power system's transient stability. This study examines a practical system of five generators and thirteen buses transient stability. The pre-fault conditions are first identified through a load flow investigation. In order to analyze post-fault conditions such, the quick fault clearing time and load switching and verify the stability of the system, fault analysis is additionally carried out. The NR approach is used for transient stability analysis, and the system's MW terminal, accelerated MW, and rotor angle are investigated for system variation under various fault conditions. PWS, or the power world simulator software is employed for conducting the simulations. It is observed that a small increase in clearing time from the critical value causes un-synchronism in the machine due to this system becomes unstable.

Key words: Power System Stability, Transient Stability, Steady State Stability, Three Phase Fault, Fault Analysis, Critical Clearing Time.

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I. INTRODUCTION

In general, a power system's three phases are generation, transmission, and distribution. While constructing a power system maintaining voltage stability in the presence of lightning, short circuits between transmission line phase wires, and ground faults is the main objective. These described issues can significantly interrupt one or more generators by producing a gap between demand and generation. [1].

As the demand in power increases, synchronization issues become more and more common during the construction of large systems. As a result, demand forecasting becomes increasingly more challenging for technological advances due to the increase in electrical power demand, which overflows the power grid. Under unexpected operation of a power system simulating a contingency situation, the voltage at each bus falls reduced due to overloading, and the generator's efficiency in providing active and reactive power decreases. To correct the problems with overhead transmission lines and increase load demand and bring the power system to safe operation, increasing power generation and constructing a new transmission line may be useful [2].

The stability of the power system is the primary issue with system operation. When constructing an electric power system, transient stability analysis is essential. The ability of a power system to withstand major interruptions such a transmission line short circuit, generator losses, and load continued operation effectively is recognized as transient stability [3]. The interruptions cause the bus voltage, power flow, and other system variables vary significantly, leading to in either a complete or partial failure of the transmission network.

Although transient stability focused into both operating environments and disturbances, steady-state stability deals with the functioning environments. The classification of power system stability becomes essential to an in-depth understanding of power reliability. The nature of system instability, the magnitude of the disturbances, and the stability in both the long and short term were thus utilized to classify the stability of the power system. The instability if the power system is a complex problem since it may happen in multiple manners and be triggered by a variety of various factors. The factors that have to be taken into consideration in addressing system instability include the physical nature of the instability, loss of synchronization, low bus voltages, high-frequency deviation, size of the disturbances, and phases created to enhance power system stability [4].

Rotor angle stability is the ability to keep synchronism after an interruption. An interruption in the system leads the synchronous machines to start "swinging" to other machines. If one machine begins operation quicker than the other and provides more loads, the deviation in angular position will increase. The system speed is decreased by increasing the speed of the other machine until balance is reached. When balance cannot be achieved, instability results, causing some machines speeds to grow until they trip [5].

Aperiodic or non-oscillatory instability, which can be brought on by a lack of synchronizing torque associated with rotor angle deviation, and periodic instability, which is carried on by a lack of damping torque associated to speed deviation, are two categories of instability. Aperiodic problems with instability are mainly brought on by insufficient synchronizing torque. As the time period is only a few seconds to a few tens of seconds, rotor angle stability is simply a short-term occurrence [6].

The large disturbance angular stability sub-section of rotor angle stability contains the power system's ability to maintain angular stability after significant disturbances, which relates to the maximum period of time that a disturbance may be provided without the system losing its stability.

Due to the highly nonlinear characteristics of transient stability, time-domain simulation is utilized to solve the algebraic and differential equations in a power system using a step-by-step calculation approach. The direct approach, such as Newton Raphson's Second approach, can also be used to determine stability. Due to the difficulty in selecting a useful stability domain and instability of finding an appropriate function, this method is not often employed [6].

Another subsection of rotor stability is the small disturbance angular stability which comprises functions of the power system and sustain stability after experiencing small disturbances. A small disturbance is considered if the linearized system still exhibits the kinetics of the original systems under this disturbance. When the system is linearized then the modal analysis method can be applied. The characteristics of modes such as mode shapes, transfer functions, and coupling coefficients are calculated by the modal analysis and are useful for improving damping. It uses the time-domain simulation to visualize the results. The branching method is another way to evaluate the effect of large changes in small-signal stability, unlike modal analysis where all system parameters are fixed [6].

Frequency stability is defined as the ability of the power system to uphold stable frequencies following disturbances due to the discrepancy between generation and load. If the system can balance or restore power and load with minimal load trips then the system is said to have frequency stability. For frequency stability analysis mainly time-domain simulations are used as an appropriate representation of dynamic devices [7].

In this study, a three-phase achieving fault is used to help in the transient stability analysis. The NR approach is used to investigate the system's swing behavior. To get accurate information of the system during its planning phase, multiple simulations are performed. The simulation of the 13-Bus system model is carried out utilizing Power Word Simulator Software. On 13-bus power systems, the work suggested is investigated under both usual and unusual operation conditions [8].

II.POWER FLOW STUDY

For the transient stability study pre-fault voltage magnitude knowledge is required. Power flow study includes the bus voltage, phase angle of the bus voltage, the transmission line actual power and reactive power, the generation bus actual and reactive power, and other variables as a piece of key information. Pre-fault conditions can be obtained by using the Newton-Raphson method from load flow analysis. Newton-Raphson method is a useful method and not affected by reserve bus selection of networking large capacity load flow solutions convergence [9]. This method starts by guessing all unidentified variables such as voltage, voltage angle of the load bus, and generator bus [10]. Firstly, the study is performed for the existence of a load bus and a distant PQ bus. For an ith bus:

$$Pi = \sum_{k=1}^{n} |Vi||Vk||Yik| \cos(\theta ik + \delta k - \delta i)\ Pi(|V|, \delta) \quad (1)$$

$$Qi = \sum_{k=1}^{n} |Vi||Vk||Yik| \sin(\theta ik + \delta k - \delta i) = Qi(|V|, \delta) \quad (2)$$

$$\text{Pi}(|\text{V}|) = \text{Pi}(\text{x}) \quad (3)$$

$$\text{Qi}(|\text{V}|) = \text{Qi}(\text{x}) \quad (4)$$

$$x = \left(\frac{\delta}{|V|}\right) \quad (5)$$

$$|\text{V}| = (|\text{V1}|, |\text{V2}|, \dots\dots, |\text{Vn}|)^t \quad (6)$$

$$\frac{\partial Pi(x)}{\partial \delta k} = -|Vi||Vk||Yik| \sin(\theta ik + \delta k - \delta i), (i \neq k) \quad (7)$$

$$\frac{\partial Pi(x)}{\partial \delta k} = \sum_{\substack{k=1 \\ k \neq i}}^{n} |Vi||Vk||Yik| \sin(\theta ik + \delta k - \delta i), (i = k) \quad (8)$$

$$\frac{\partial Pi(x)}{\partial |Vk|} = |Vi||Vk||Yik| \cos(\theta ik + \delta k - \delta i), (i \neq k) \quad (9)$$

$$\frac{\partial Pi(x)}{\partial |Vk|} = 2|Vi||Yii| cos\theta ii + \sum_{\substack{k=1 \\ k \neq i}}^{n} |Vk||Yik| \cos(\theta ik + \delta k - \delta i), (i = k) \quad (10)$$

$$\frac{\partial Qi(x)}{\partial \delta k} = |Vi||Vk||Yik| \sin(\theta ik + \delta k - \delta i), (i \neq k) \quad (11)$$

$$\frac{\partial Qi(x)}{\partial \delta k} = \sum_{\substack{k=1 \\ k \neq i}}^{n} |Vi||Vk||Yik| \cos(\theta ik + \delta k - \delta i), (i = k) \quad (12)$$

$$\frac{\partial Qi(x)}{\partial |Vk|} = |Vi||Vk||Yik| \sin(\theta ik + \delta k - \delta i), (i \neq k) \quad (13)$$

$$\frac{\partial Qi(x)}{\partial |Vk|} = 2|Vi||Yii| sin\theta ii + \sum_{\substack{k=1 \\ k \neq i}}^{n} |Vk||Yik| \sin(\theta ik + \delta k - \delta i), (i = k) \quad (14)$$

III.METHODOLOGY

Transient stability addresses the effects of accidental outages, fault incidences, abstraction or application of sudden loads. the practical system is used load flow is completed on the system model. Load flow analysis was performed using adaptive newton Rapson method. After load flow analysis the voltages at a bus, MW at load and generator, MVAR at load and generator as shown.

TABLE I. Load Flow Analysis Results Without Fault on the 13-Bus System

Bus no.	Bus type	Voltage (PU)	Voltage (KV)	MW Load	MVAR Load	MW Generator	MVAR Generator
1	Slack bus	1.04	16.5	0	0	0	0
2	PV	1.025	18	163	6.7	0	0
3	PV	1.025	13.8	85	-10.9	0	0
4	PQ	1	230	0	0	0	0
5	PQ	1	230	0	0	125	50
6	PQ	1	230	0	0	90	30
7	PQ	1	230	0	0	0	0
8	PQ	1	230	0	0	100	35
9	PQ	1	230	0	0	0	0
10	PQ	1.06	230	0	0	0	0
11	PV	1	230	40	30	20	10
12	PQ	1	230	0	0	45	15
13	PQ	1	230	0	0	40	5

Table II. Line Data For 13-Bus

SL. No.	From bus	To bus	R	X	B
1	1	4	0	0.0576	0
2	4	5	0.01	0.85	0.176
3	4	6	0.017	0.092	0.158
4	6	9	0.039	0.17	0.358
5	5	7	0.032	0.161	0.306
6	9	3	0	0.0586	0
7	7	2	0	0.0625	0
8	9	8	0.0119	0.1008	0.209
9	7	8	0.0085	0.072	0.149
10	10	11	0.02	0.06	0.03
11	10	12	0.08	0.24	0.025
12	11	12	0.06	0.18	0.02
13	11	13	0.06	0.18	0.02
14	12	13	0.01	0.03	0.01

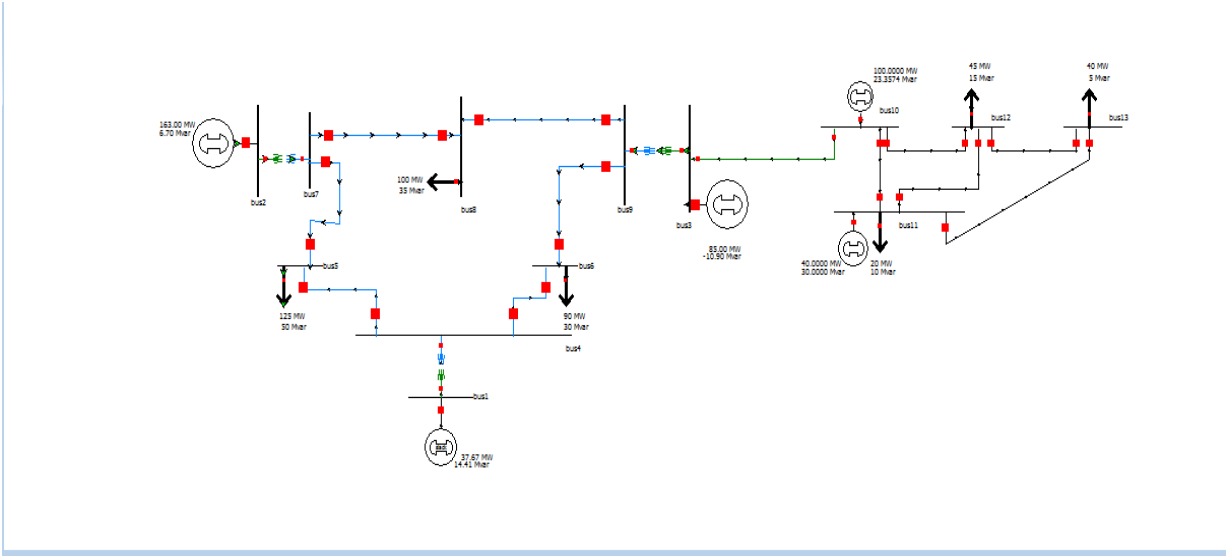


Figure 1. Practical 13-Bus System

IV. RESULT AND DISCUSSION

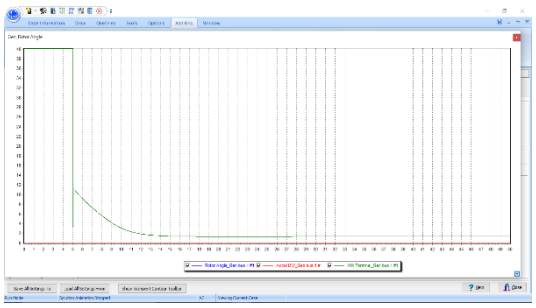


Figure 2. MW terminal v/s time, fault at bus 7

Fault occurring time of 5 sec and clearing time of 5.05 sec.

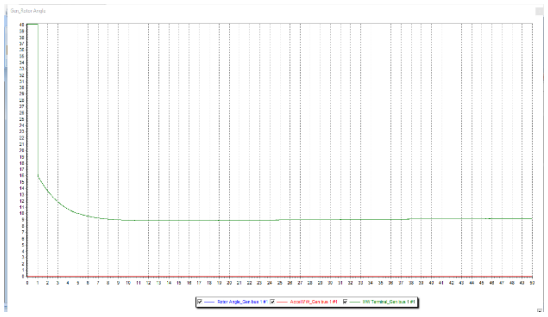


Figure 3. MW terminal v/s time, fault at bus 11

Fault occurring time of 1sec and clearing time of 5 sec. Hence the system is Stable system.

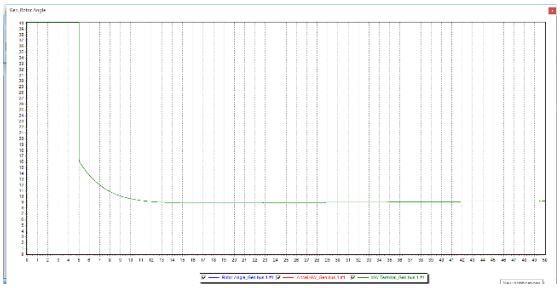


Figure 4. MW terminal v/s time, fault at bus 11

Fault occurring time of 5 sec and clearing time of 5.05 sec. Hence the system is Stable system.

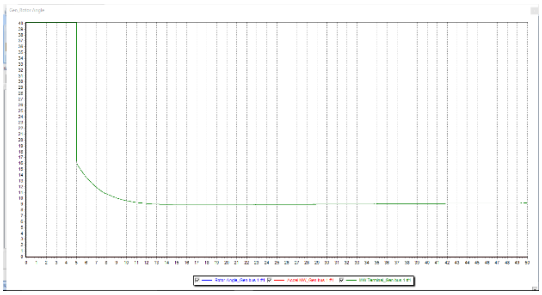


Figure 5. MW terminal v/s time, fault at bus 11

Fault occurring time of 5 sec and clearing time of 8 sec. Hence the system is Stable system

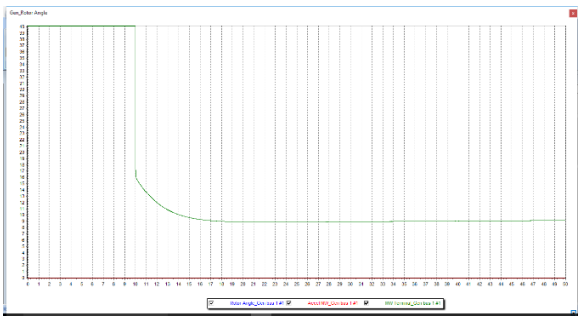


Figure 6. MW terminal v/s time, fault at bus 11Fault occurring time of 10 sec and clearing time of 11 sec. Hence the system is Stable system.

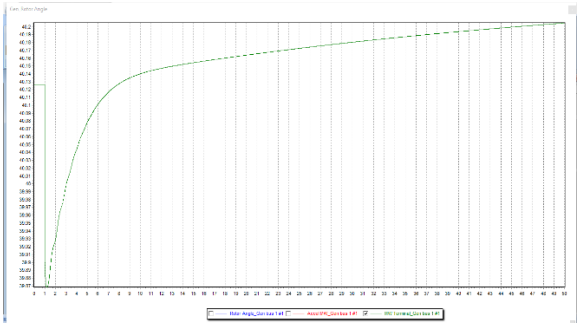


Figure 7. MW terminal v/s time, fault at bus 11

Fault occurring time of 1 sec and clearing time of 2000 sec. Hence the system is Unstable system.

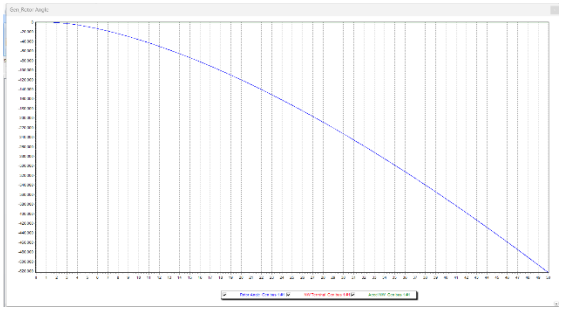


Figure 8. Rotor Angle v/s time, fault at bus 4

Fault occurring time of 1 sec and clearing time of 5 sec. Hence the system is Stable system.

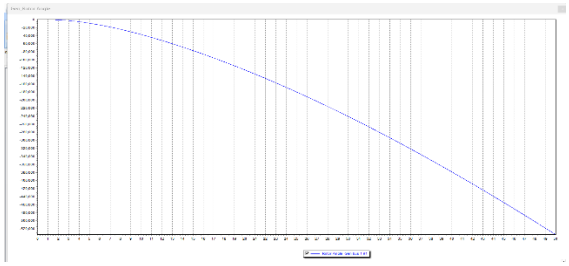


Figure 9. Rotor Angle v/s time, fault at bus 4

Fault occurring time of 1 sec and clearing time of 10 sec. Hence the system is Stable system.

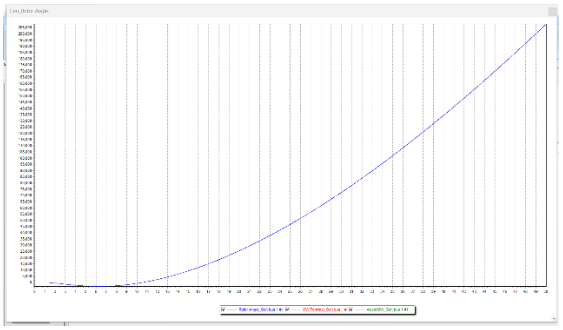


Figure 10. Rotor Angle v/s time, fault at bus 4

Fault occurring time of 1 sec and clearing time of 1.5 sec. Hence the system is Unstable system.

V.CONCLUSION

Using the power world simulator software, various fault types have been simulated on the practical 13-bus system in order to investigate the transient stability of the power system. The NR approach was used for analyzing the load flow for this purpose. It was determined through transient stability investigations that increasing CCT increases the rotor angle difference. Additionally, the results of multiple investigations of fault impact on the power system illustrate that, although fault clearness before CCT maintains the system's stability, fault clearness after CCT makes the system unstable.

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